

Early signals of breakdown through Stochastic modeling

Yinon Ashkenazy & Michael Assaf
Racah Institute of Physics,
Hebrew University, Jerusalem, Israel



האוניברסיטה העברית בירושלים
The Hebrew University of Jerusalem

Dislocation mediated – self organized criticality

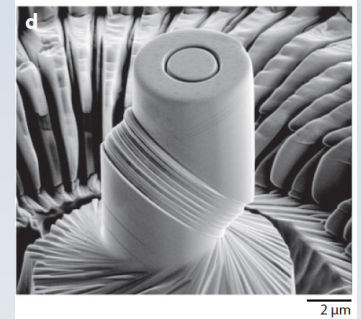
Plasticity of Micrometer-Scale Single Crystals in Compression

Michael D. Uchic,¹ Paul A. Shade,²
and Dennis M. Dimiduk¹

Single crystal micro-pillar compression:

Dislocation mediated intermittent flow – size effects, hardening.

Dislocation density inside a plane as a controlling parameter.



Direct quantitative analysis of strain bursts (~20 micron).

Intermittency characterized by a universal Power law burst PDF

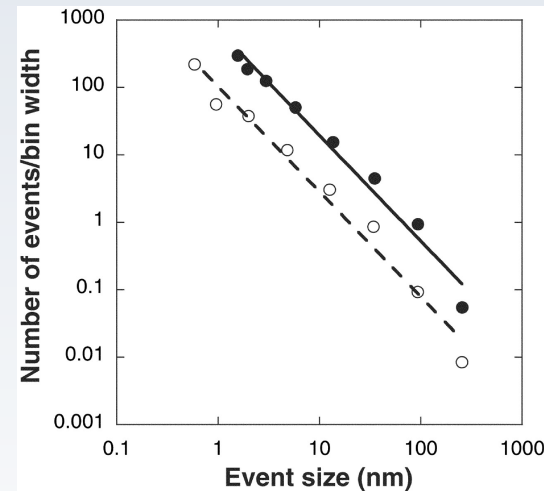
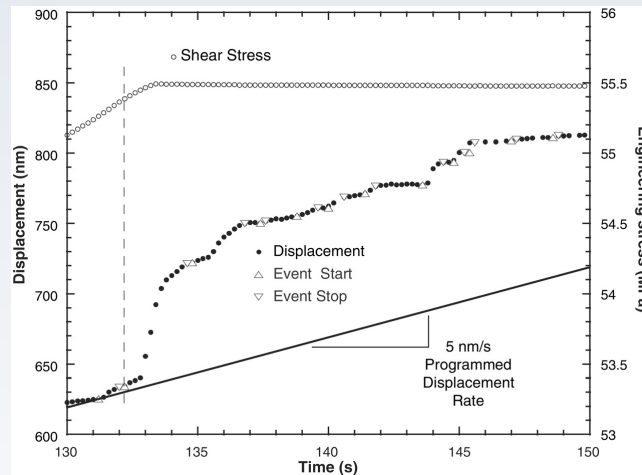
Acoustic emissions:

Similar + space and time coupling between events

(Weiss & Marsan, Science 2003)

Earthquakes show similar PDF and spatio-temporal correlation

(Kagan, Geophysical J. (2007))

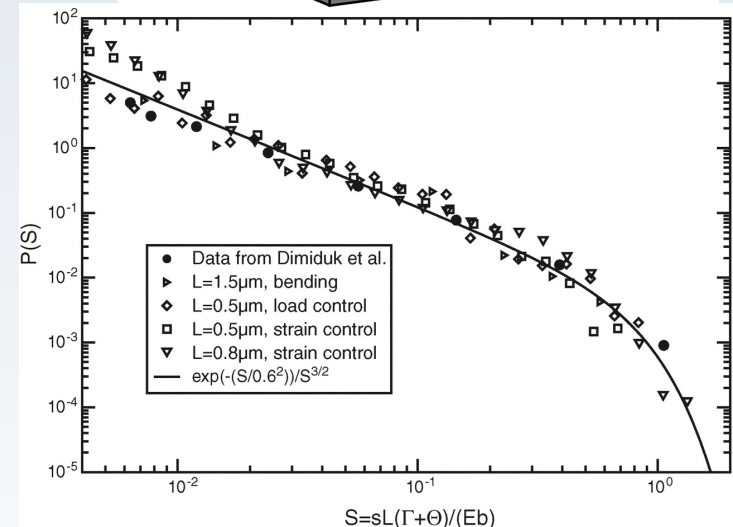
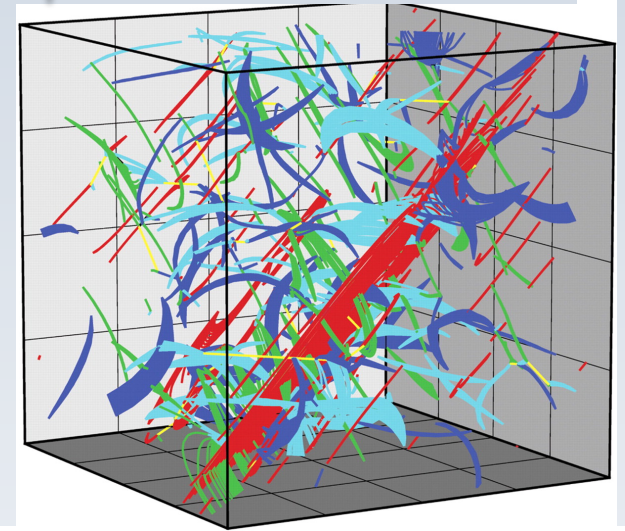


Uchic, Shade & Dimiduk, Annual Review of Materials Research (2009).

Dimiduk, Woodward, LeSar & Uchic: "Scale-Free Intermittent Flow in Crystal Plasticity." Science (2006) 1188.

Using dislocation dynamics to reproduce PDF

- 3D dislocation dynamics reproduce strain burst scaling $P(s) = Cs^{-\tau} \exp\left[-(s/s_0)^2\right]$
- where C is a normalization constant, τ is a scaling exponent, and s_0 is the characteristic strain of the largest avalanches.
- Intermittency – as a result of dislocation Interactions. Stochastic nature a result of varying initial conditions.
- Avalanche is a 2D event, with an upper cutoff due to structure and work-hardening. Strain is limited to about 10^{-6} in a cm size sample.
- Recently (Chen, choi, papanikolaou & Sethna 2010 to 2013): scaling of structures using an advanced CDD code.

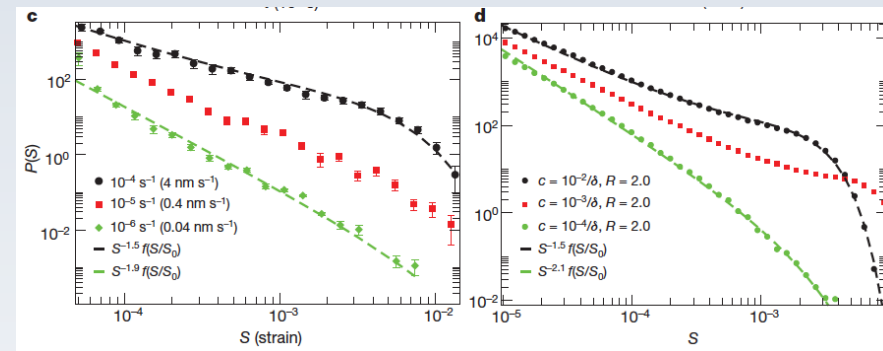


Mean field models for critical depinning

Quasi-periodic events in crystal plasticity and the self-organized avalanche oscillator

Stefanos Papanikolaou¹, Dennis M. Dimiduk², Woosong Choi³, James P. Sethna³, Michael D. Uchic², Christopher F. Woodward² & Stefano Zapperi^{4,5}

- Reproduce strain rate variation by modifying the mean field picture to include a competing relaxation mechanism. This lead to oscillation in avalanche size. (nature, 2012)



PRL **109**, 105702 (2012)

PHYSICAL REVIEW LETTERS

week ending
7 SEPTEMBER 2012

Distribution of Maximum Velocities in Avalanches Near the Depinning Transition

Michael LeBlanc,¹ Luiza Angheluta,^{1,2} Karin Dahmen,¹ and Nigel Goldenfeld¹

- Using a mean field model for interface depinning and by solving Fokker-Planck eq. reproduced the power law decay of avalanche size and maximal velocity

$$\frac{dV}{dt} = -kV + F_c + \sqrt{V}\xi(t)$$



What are we trying to do...

- Use stochastic theory to allow for:
 - transferability of failure scenario analysis (across drive conditions)
 - Identify controlling mechanisms
 - Define critical experiments - model development / verificationSuch models serve as a link between the microscopic, short time scale problem which is accessible via simulation to the measured system to the real life scenario.

For now – demonstrate the basic method using a “spherical horse” model.

Not trying to do (at this stage):

Create a comprehensive consistent microscopic model

Describe the “real” mechanism at work

Solve the full field - structure - current response function

DC vs RF



Defect model for the dependence of breakdown rate on external electric fields

K. Nordlund and F. Djurabekova

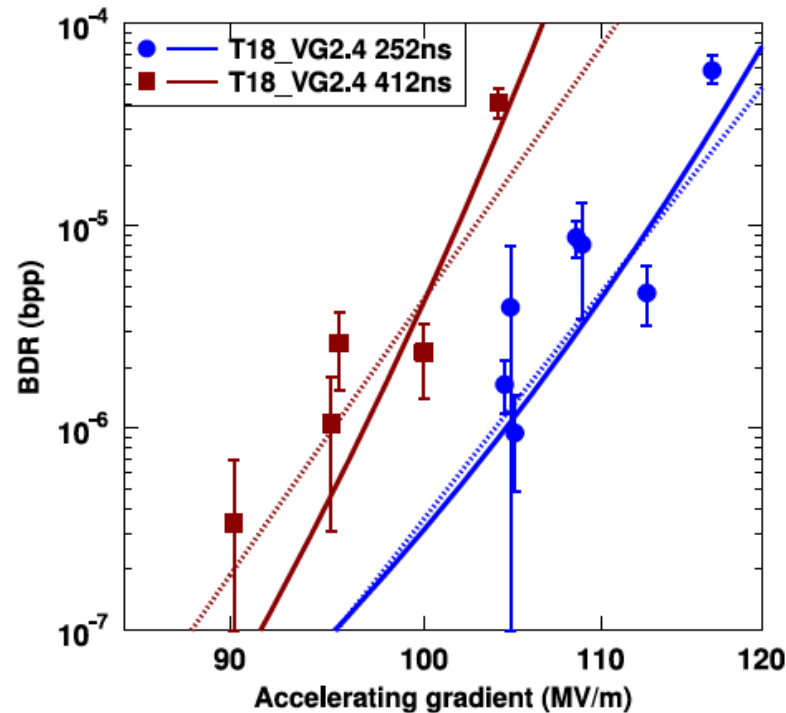


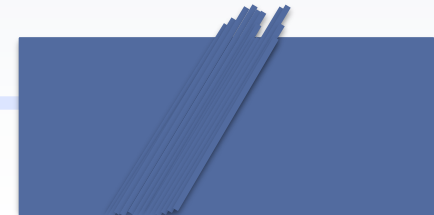
FIG. 4. Measured dependences of R_{BD} (in units of breakdown per pulse, bpp) versus electric field for the T18 accelerating structure [33,43] and fits of our model (solid lines) as well as power laws (dashed lines) to the data.

$$\frac{1}{\tau_{RBD}} \approx \exp\left[E^2 \Delta V / (k_B T)\right]$$



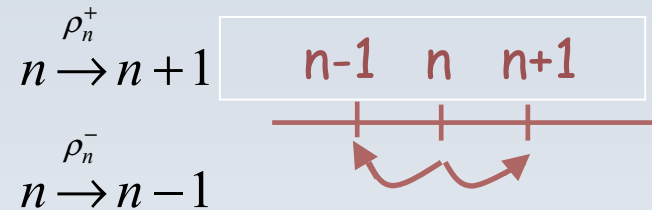
Formulation of a “well-mixed” 0d model

- Assumptions:
 - Breakdown currents are driven by formation of surface extrusion/intrusions.
 - Surface protrusions are formed due to multiple dislocation reaction leading to local geometric features
 - Sub-breakdown surface protrusion are not identified (true?).
Breakdown rates do not go up with time (BDR even goes down...).
Therefor we assume that gradual protrusion accumulation does not control breakdown:
 - surface relaxation, interaction between various slip systems, protrusion-dislocation interaction...
 - Field conditions are translated to an applied stress
(AC thermal gradients ~ 100 Mpa, dc?)
- Suggested controlling parameter –
the number of mobile dislocations inside a band.
 - If large amount of dislocations reach the surface in unison – an instant extrusion/intrusion may lead to breakdown.
 - We avoid spatial interaction and assume gain-loss dynamics inside a specific band.



General gain-loss type Markovian processes

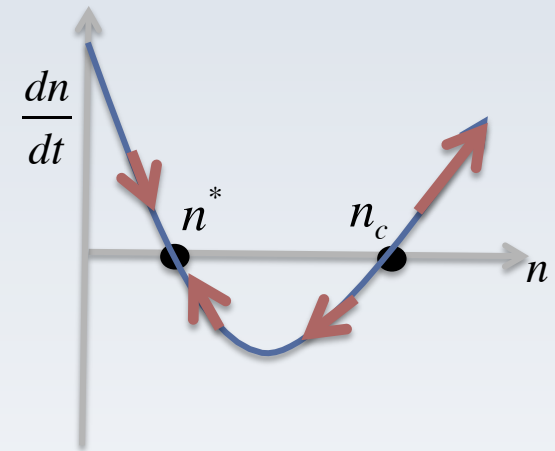
Rates for transition between states



The master equation

$$\dot{P}_n = \rho_{n-1}^+ P_{n-1} + \rho_{n+1}^- P_{n+1} - (\rho_n^+ + \rho_n^-) P_n$$

can lead to bifurcation:
a metastable state and a critical one.



We look for the quasi-stationary probability distribution function
And the probability to cross the critical point (reach extinction)

Approximate solution based on WKB theory with $1/N$ being the small parameter.

$$\dot{P} = 0 \Rightarrow P(n) \equiv P(\rho N) \sim e^{-N[S(\rho) + O(1/N)]}$$


“Minimal” model

- Define the “in-plane” density (in units of 1/nm).
- External stress (due to temp gradient on surface), range of 0.1 Gpa.
- Mobile dislocations can increase in number due to stress gradient (the driving force) as well as thermal activation of the multiplication reaction

$$\frac{d\rho^+}{dt} = v_0 G A (\sigma(\rho))^k e^{-(\phi_0 - \sigma\Omega)/k_B T}$$
- Moving dislocations can become sessile at:
 - Pre-existing barriers (concentration - C)
 - “collisions” with other moving dislocations

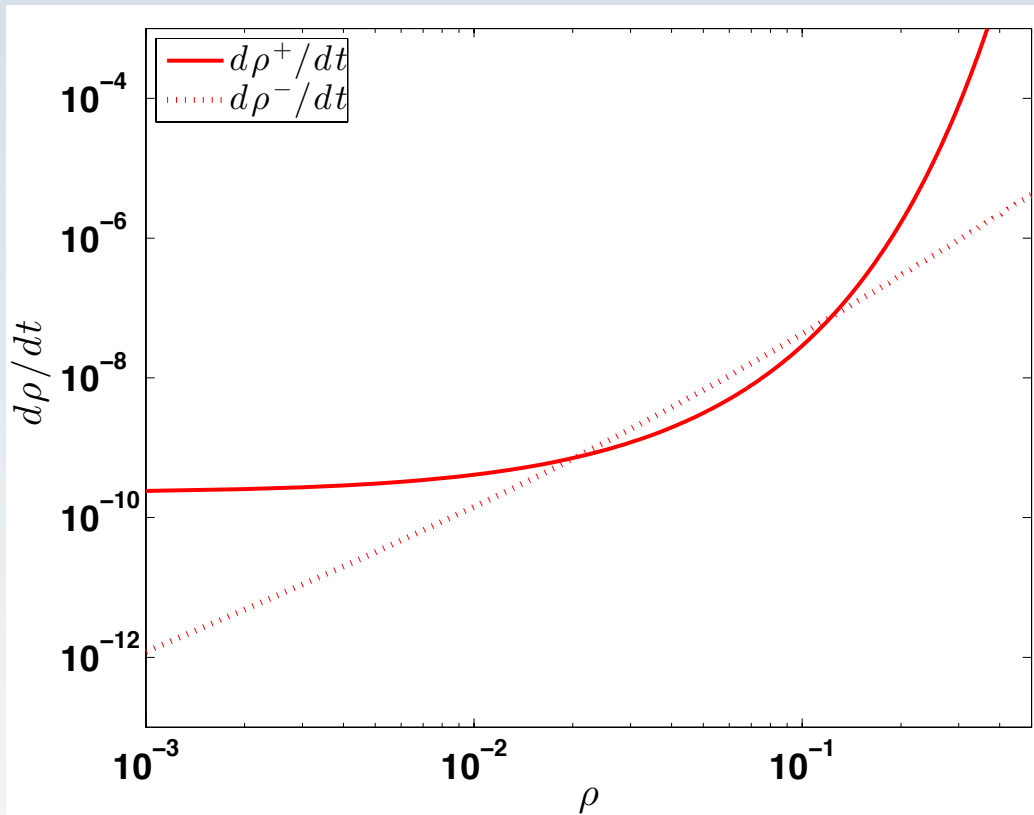
$$\frac{d\rho^-}{dt} = \rho V(\sigma) c + \rho 2V(\sigma) \rho$$
- Properties dependence:
 - Velocity increase with stress, independent of the number of moving dislocations
 - Stress increase with dislocation content

$$\sigma = \sigma_E + G\rho b$$

$$V = Bv_0 b^2 \sigma$$

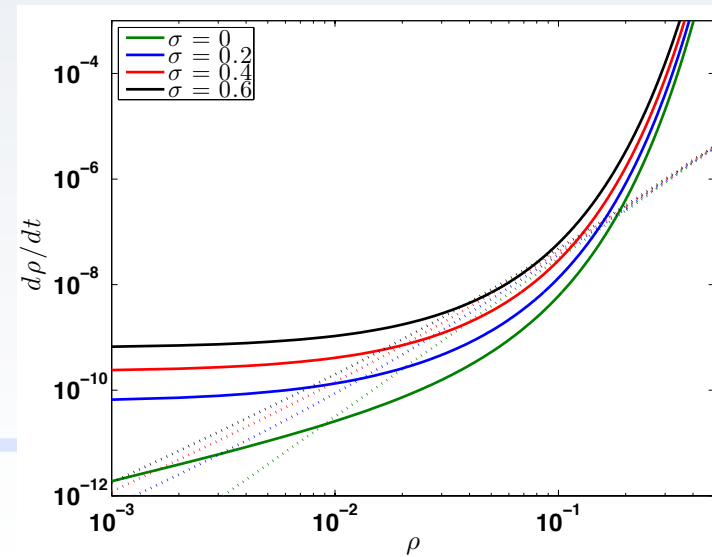


Model characteristics

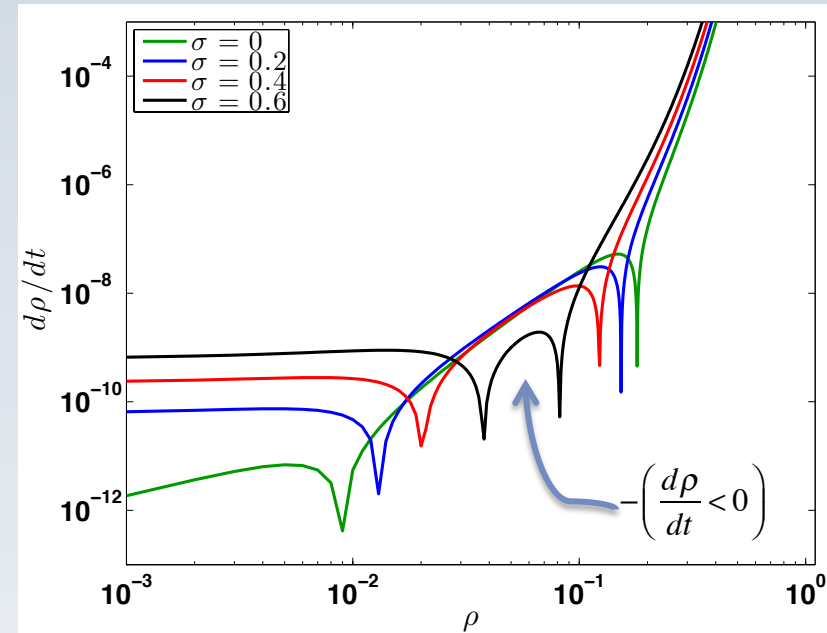


Stable point:
Dislocation generation and annihilation identical

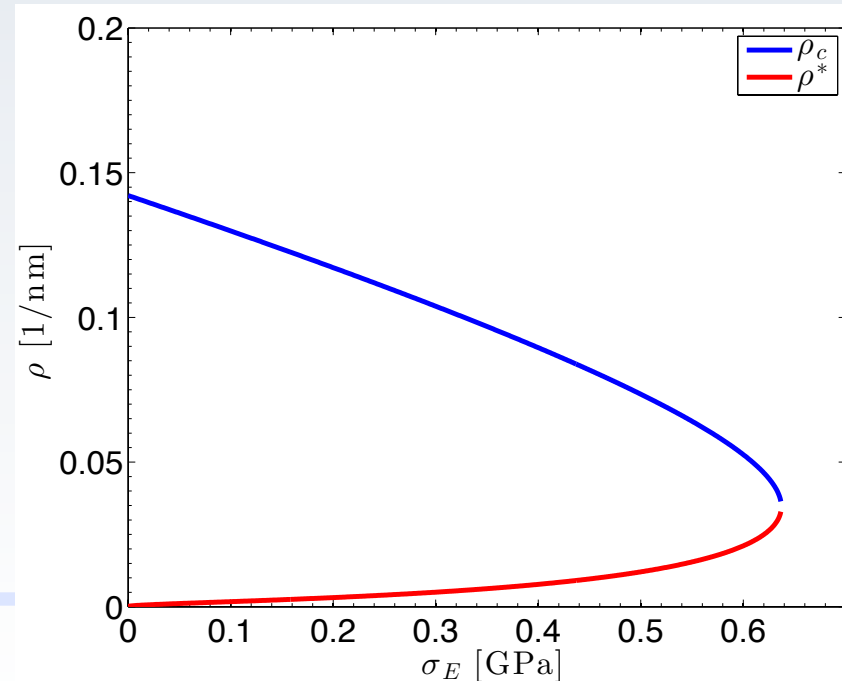
Critical point – threshold for dislocations avalanche



Low stresses:
Mobile dislocation density remains in
Metastable solution.
Dynamic barrier decreases with increasing
stress.



Up to a critical stress ρ bifurcation to two
solutions.
Above it – no stable solution.



Define :
$$s(\rho) = - \int_{\rho} \ln \frac{\rho^+(x)}{\rho^-(x)} dx$$

For (k=2):
$$s(\rho) = -\rho \ln \left[\frac{\rho^+(\rho)}{\rho^-(\rho)} \right] - \frac{\sigma_E}{b\mu} \ln [\sigma(\rho)] + \frac{c}{2} \ln(c + 2\rho) - \rho \left(1 - \frac{\rho b \mu \Omega}{2k_B T} \right)$$

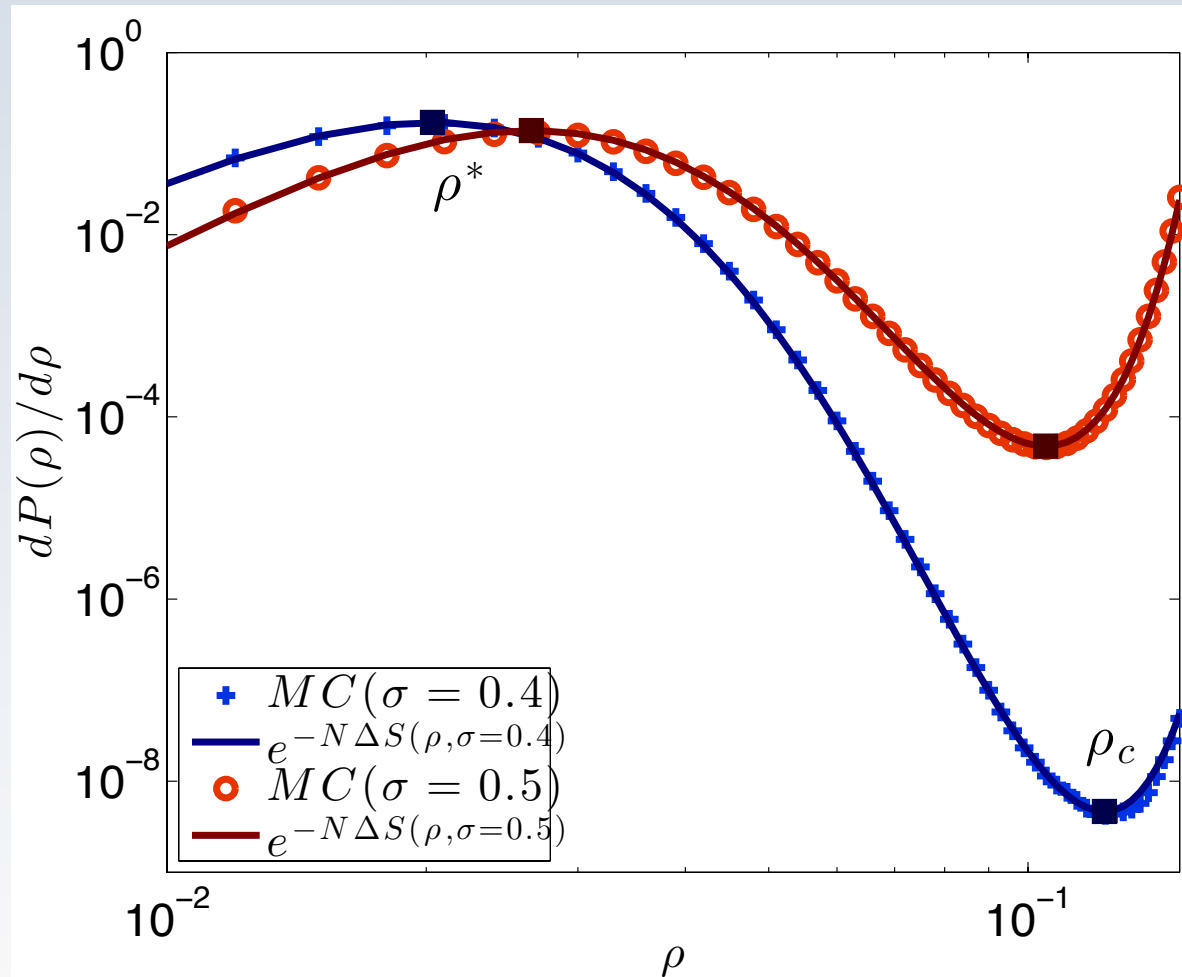
Using*:
$$P(n = N\rho) = P(\rho) \sim e^{-Ns(\rho)}$$

Leads to
$$P(\rho) \propto \left[\frac{\rho^+(\rho)}{\rho^-(\rho)} \right]^{N\rho} \frac{[\sigma(\rho)]^{N\sigma_E/b\mu}}{(c + 2\rho)^{Nc/2}} e^{N\rho \left(1 - \frac{\rho b \mu \Omega}{2k_B T} \right)}$$

And the normalized PDF
$$P(\rho) = \underbrace{\sqrt{\frac{S''(\rho^*)}{2\pi N}}}_{P(\rho^*)} e^{-N[s(\rho) - s(\rho^*)]}$$



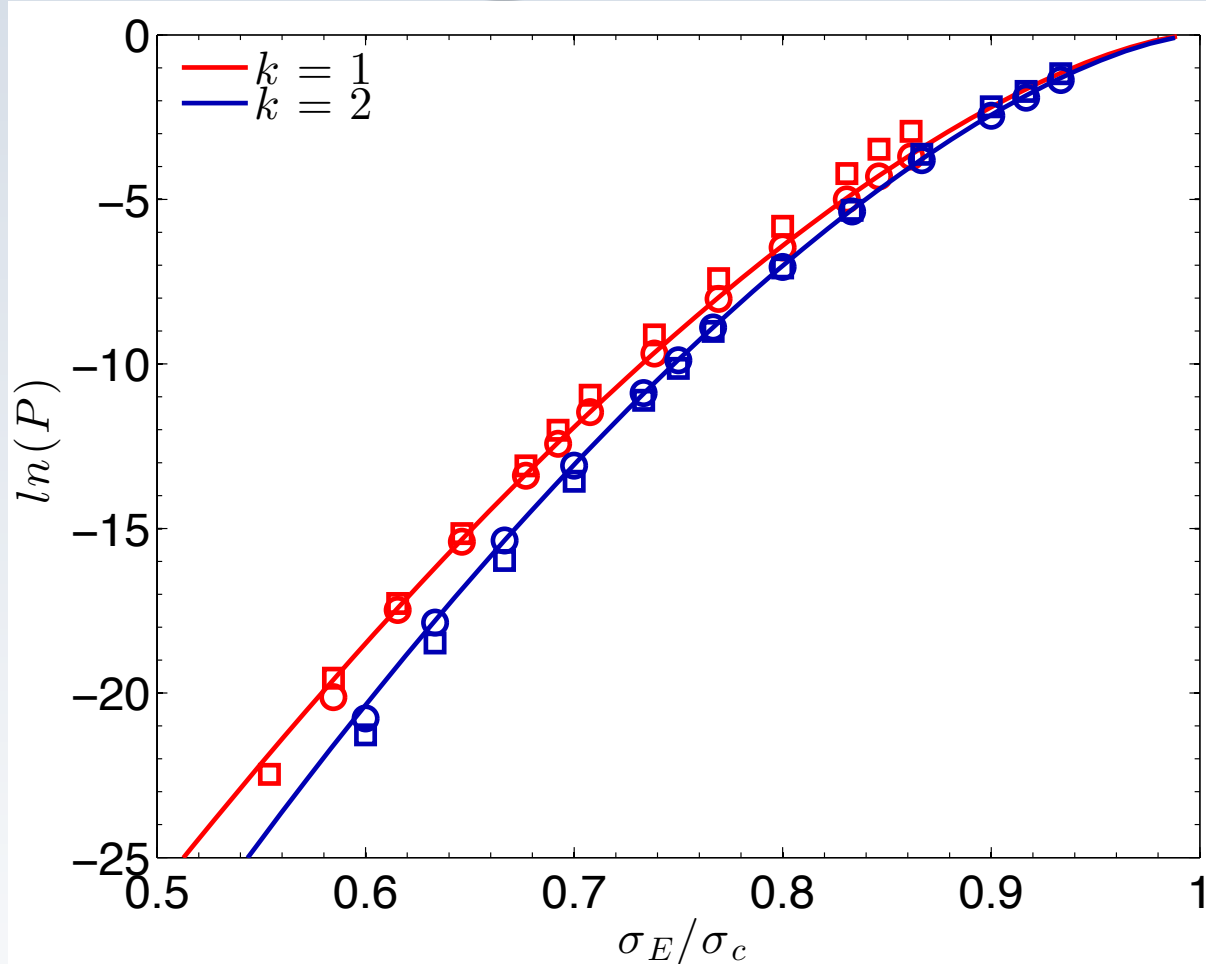
Analytical vs MC



- Analytical analysis reproduces full PDF.



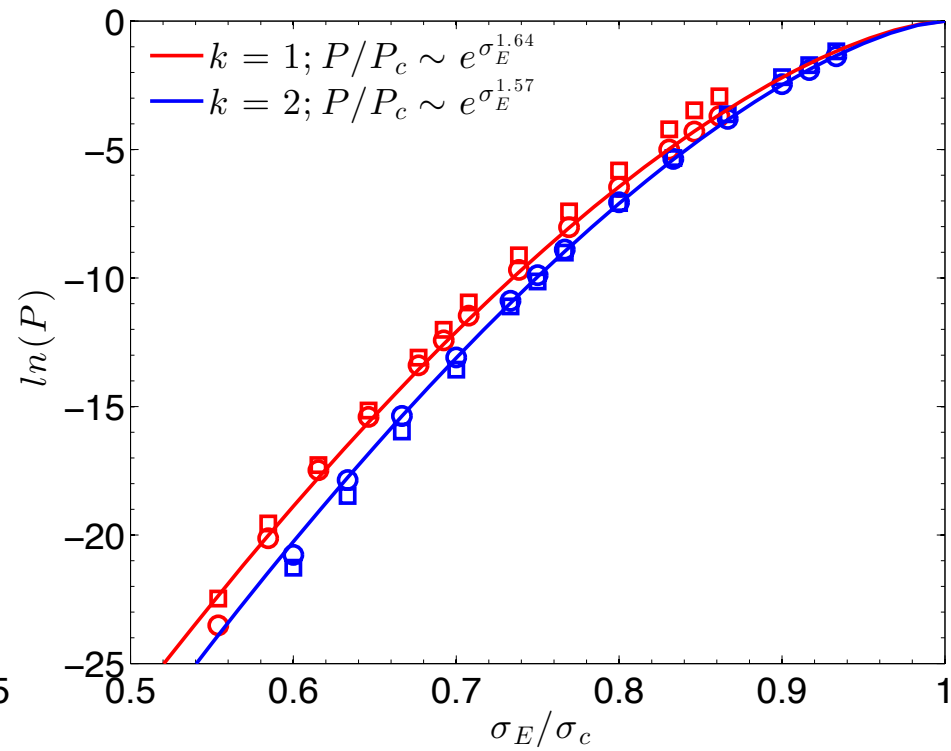
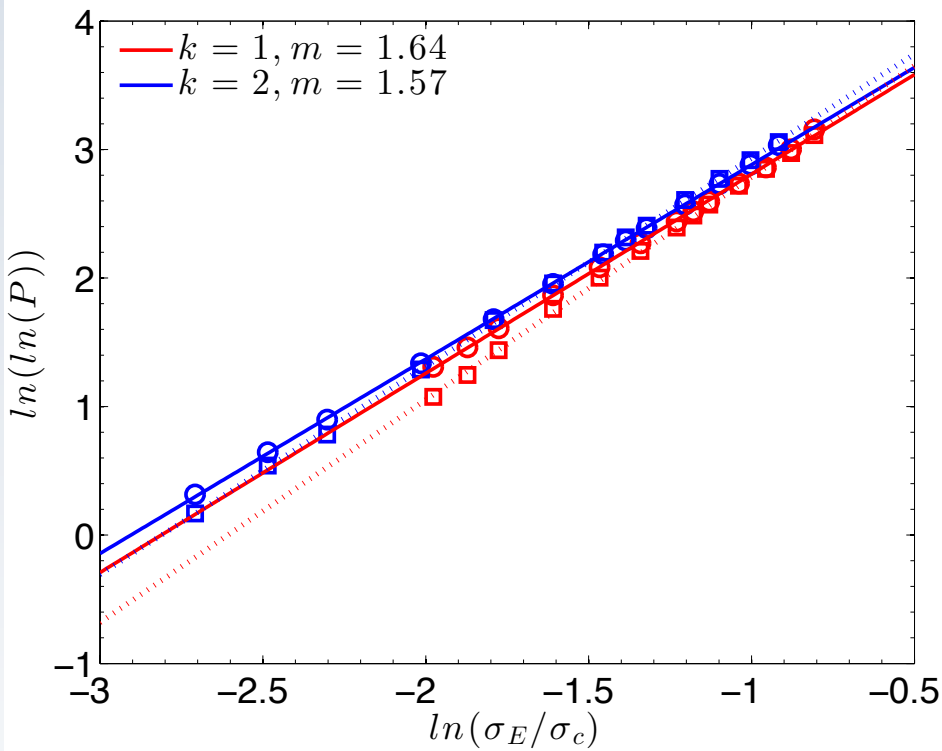
Getting BDR curves



- Analytical solution for relative probability to reach critical point.
- Normalized probability and rate for reaching the critical state.



“Universal” BDR exponent



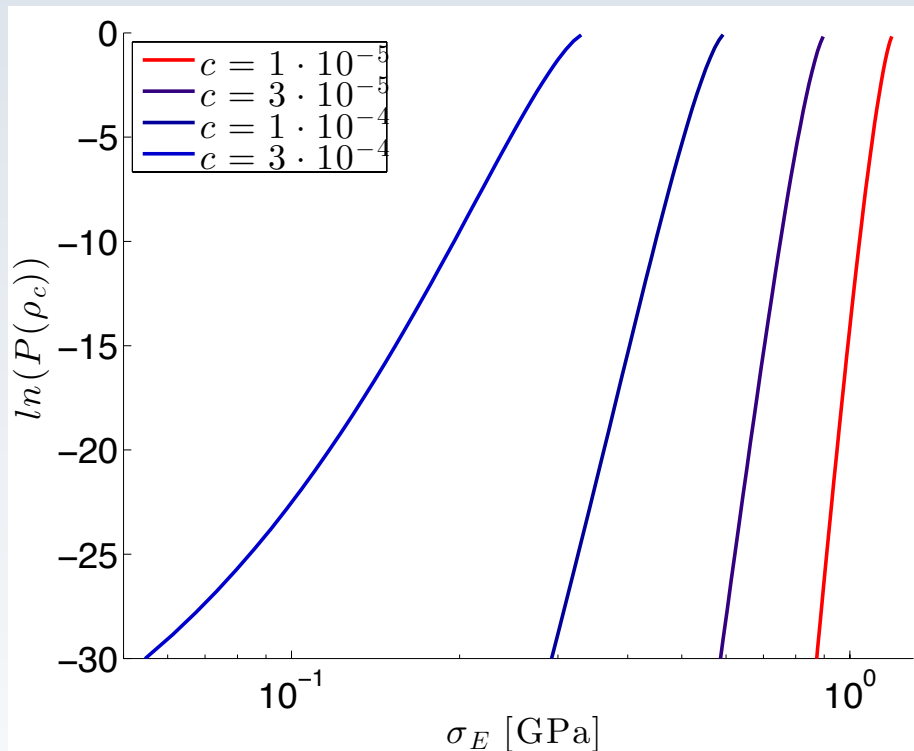
- fit to experimental results

$$P \sim \frac{1}{\tau} \sim \exp(\sigma_E^{1.6})$$

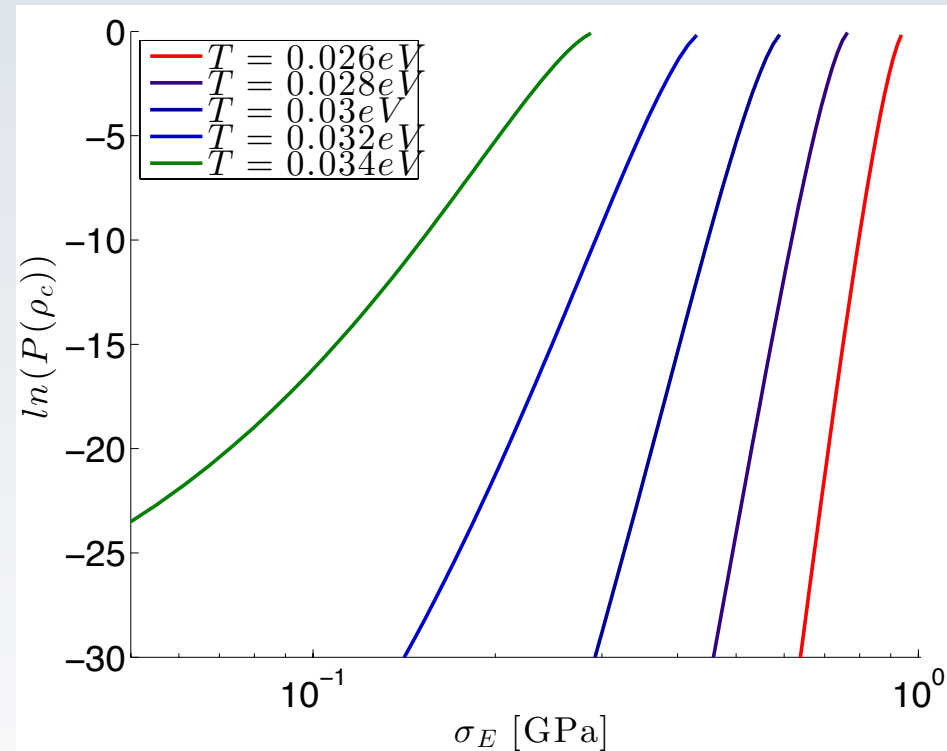


Other dependencies

Employ analytical solution to various scenarios



Dependency on mobile dislocation generation pre-factor



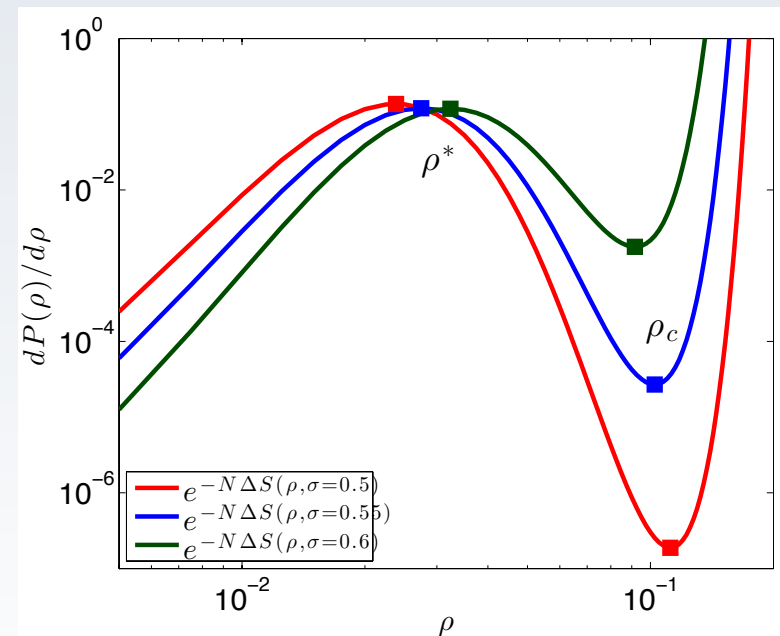
Temperature dependence.



Signs of criticality

- Adiabatically moving between quasi-stationary PDF:
Change in pdf moments with field
→ identify threshold
- At specific conditions, probe time dependencies of the QS pdf:
Identify large fluctuations time dependency
→ identify time constants
→ mechanism

$$P_c(\sigma, \rho, t) = \int_0^t P(\sigma, \rho' > \rho, t') dt'$$



PRE-breakdown

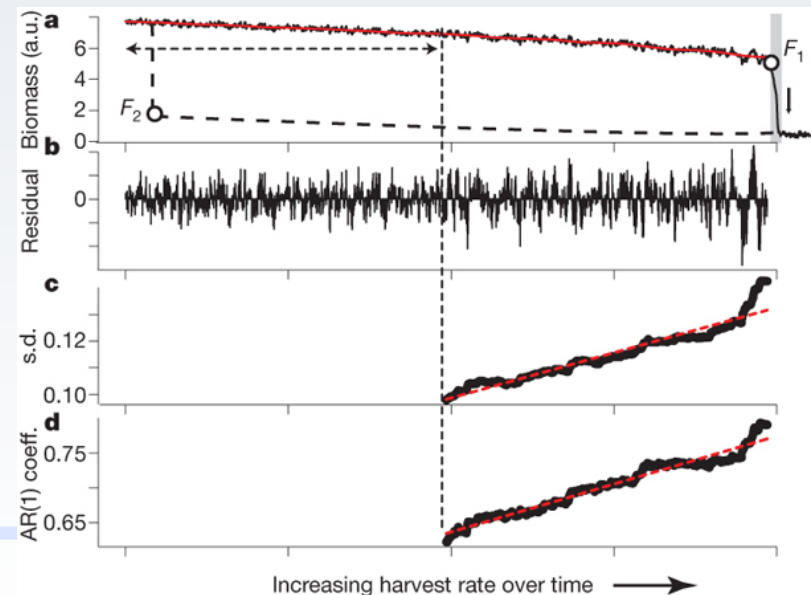
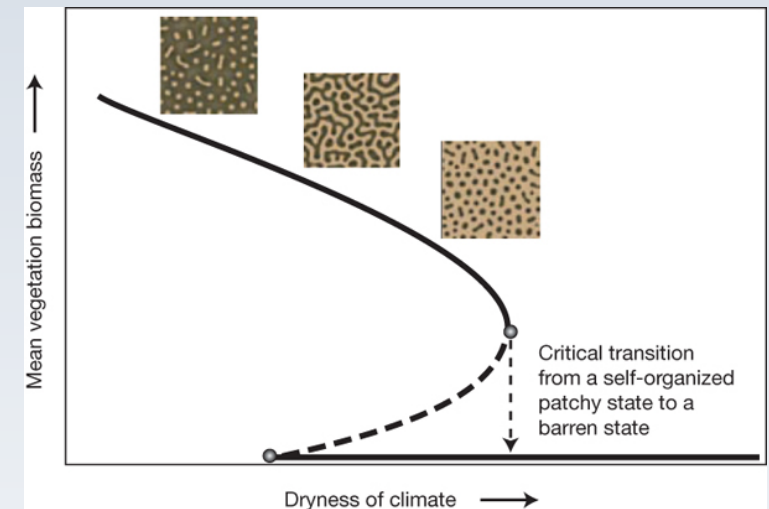
- As the system approaches the critical point. Fluctuation diverge.
- Observable through standard deviation of the time correlation

$$SD(t) = \frac{\int_{t-\Delta}^{t+\Delta} (I(t) - \langle I \rangle)^2 dt}{(\langle I \rangle)^2}$$

- Or, more generally, autocorrelation in the signal
- $$R(k) = \frac{\int_0^{t-k} (I(t) - \langle I \rangle)(I(t+k) - \langle I \rangle) dt}{\int_0^{t-k} (I(t) - \langle I \rangle)^2 dt}$$

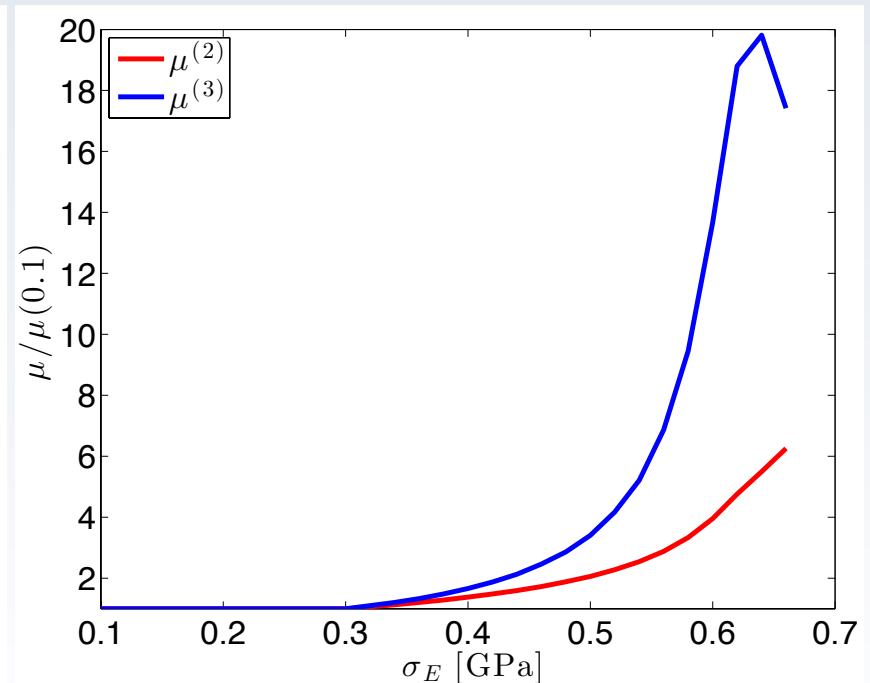
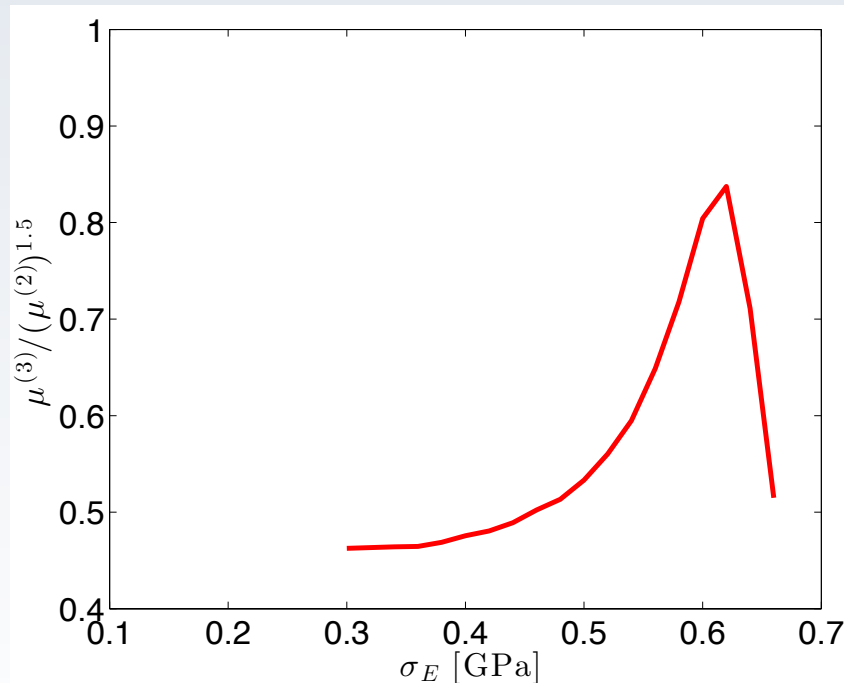
Early-warning signals for critical transitions

Marten Scheffer¹, Jordi Bascompte², William A. Brock³, Victor Brovkin⁵, Stephen R. Carpenter⁴, Vasilis Dakos¹, Hermann Held⁶, Egbert H. van Nes¹, Max Rietkerk⁷ & George Sugihara⁸



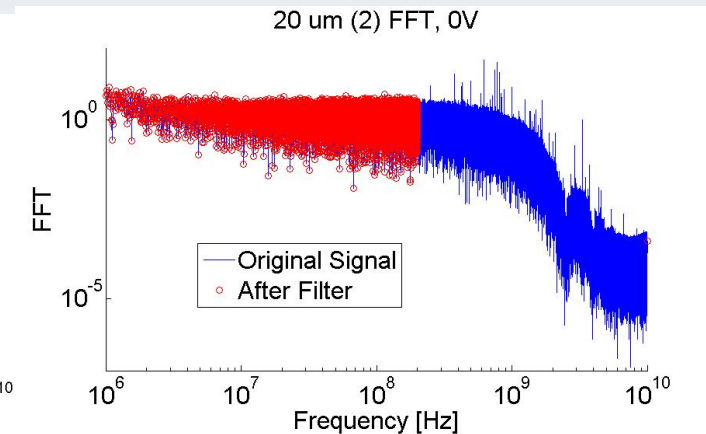
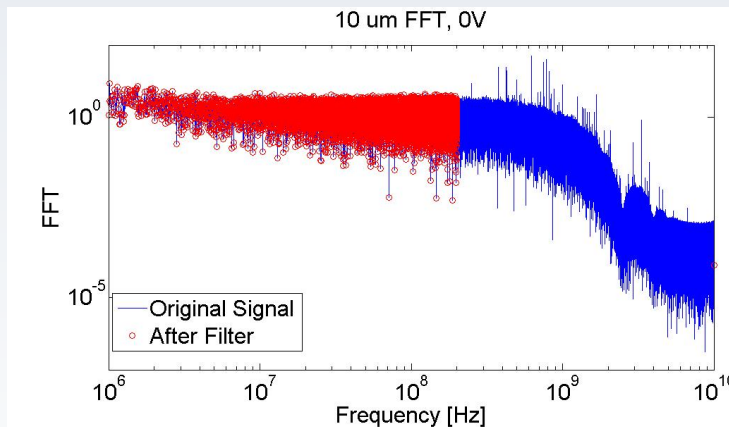
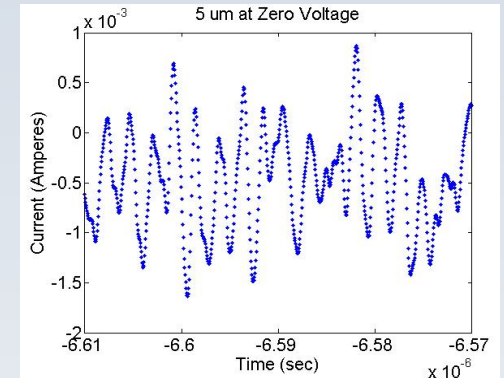
Early warning signals

- Using moments $\mu^{(n)} = \sum (\rho - \langle \rho \rangle)^n$
- Variance and third moment indicate to Strong variation of quasi-stationary PDF – in breakdown states



Initial try – DC measurements

- Dark current measurements
(Varying field and gap distance)
- Low pass filter is clearly needed
(applied 0.2 GHz)

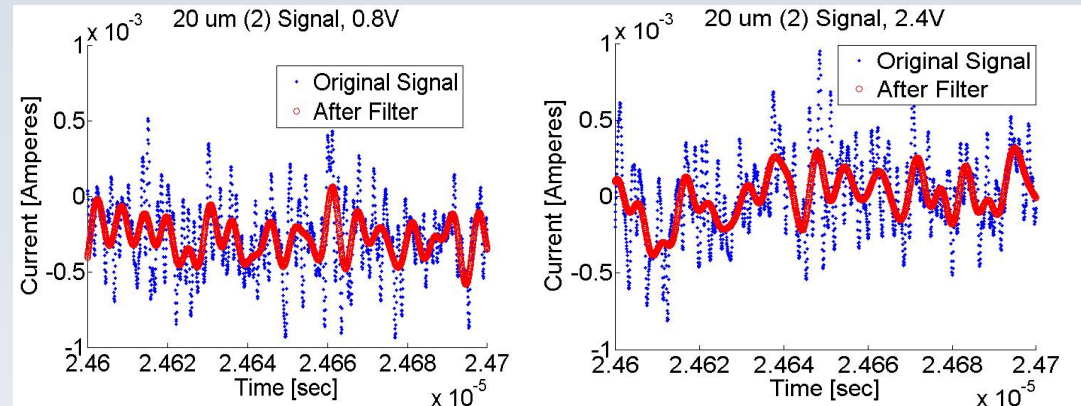


Nick Shipman, Adar Sharon

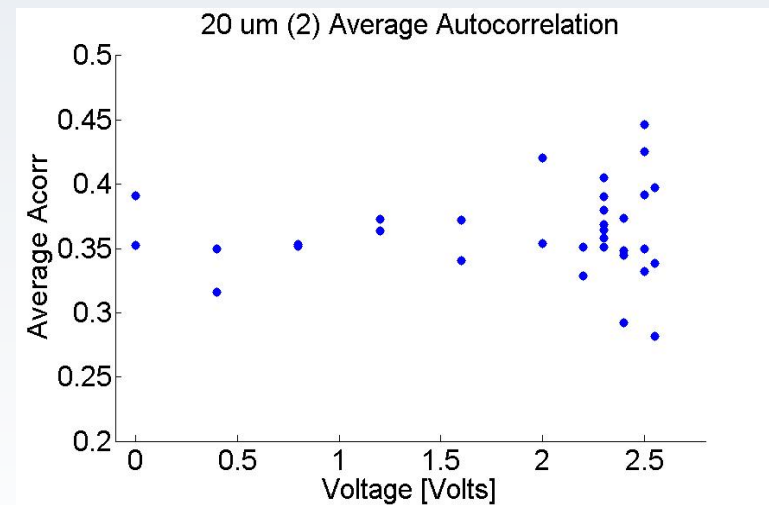
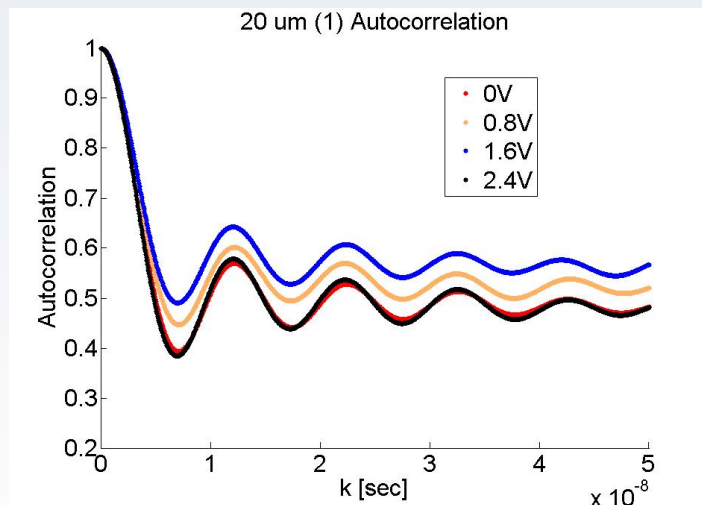


Initial try – DC measurmanets

- Filtered signal

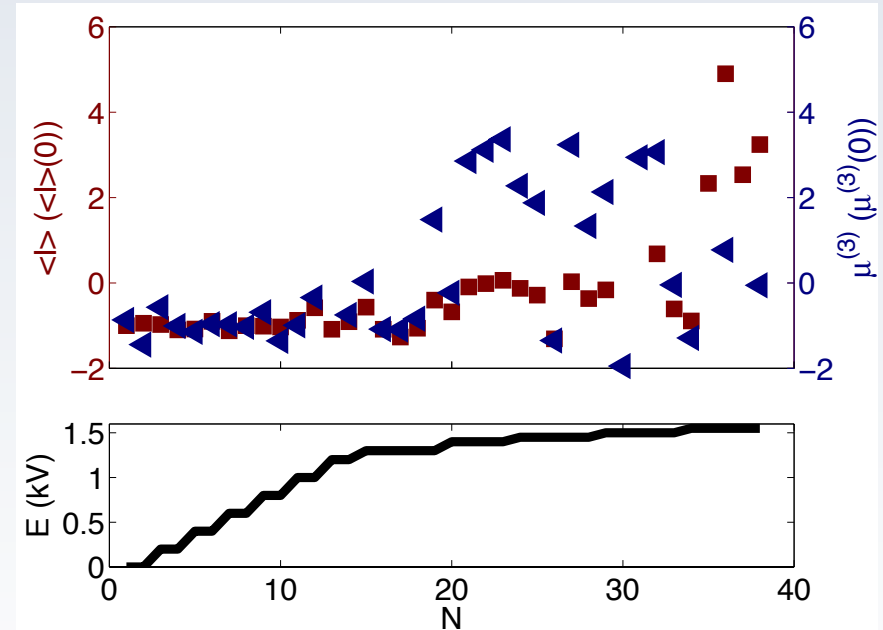
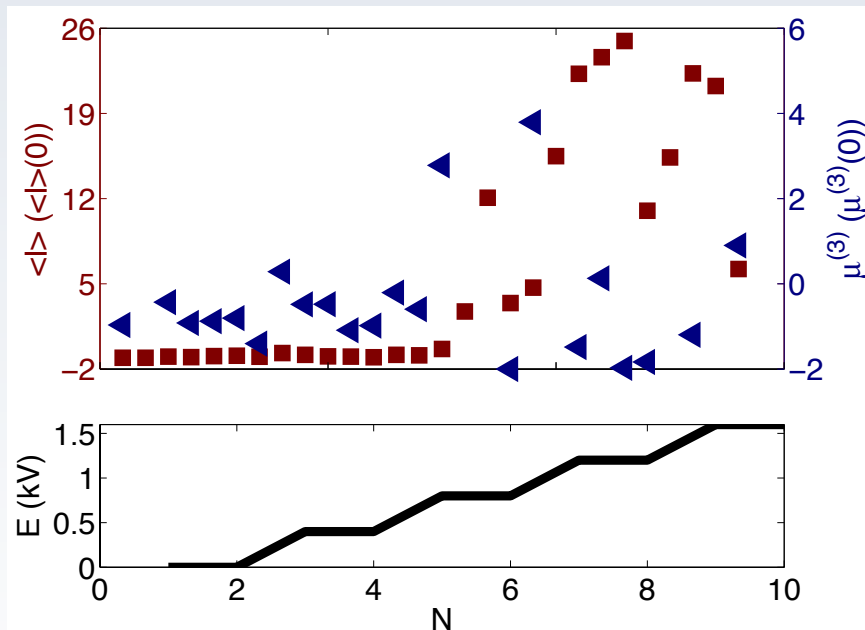


- Sadly ... autocorrelation shows nothing...



Normalized average and third moment

- Mean and third moments variations with field.
- Need for better measurements ...



Summary

- Intermittency due to collective dislocation response is well established.
 - Experimental scenarios: acoustic emission, micro-compression.
 - Universal behavior, – earth quakes, other non local bifurcating systems.
- Proposed a simple stochastic model to describe breakdown phenomena
 - Using a minimal model – MANY simplifying assumption – demonstrate critical behavior, bifurcation and reproduce observed BDR (E) .
 - Analytically (or at least numerically) solvable
 - reproduce observed BCR exponent. Universality?
- Unique experimental scenarios:
 - PDF – pre breakdown analysis: using PDF and PDF tail.
 - Pre breakdown fluctuation.
- Future directions
 - Effects of intrinsic noise?
 - Join forces with experimental work to identify signs of criticality.
 - Suggest a modified conditioning scheme .
- Can serve to bridge microscopic mechanisms to experimental scenarios. – need for an explicit response function.
- New opportunity for stochastic analysis...

